

Part 1: Classful Addressing

1. Open Cisco Packet Tracer:

- Double-click the **Cisco Packet Tracer** icon on your computer to open the software.

2. Create a Network Topology:

- In the left pane, select the **End Devices** category and drag two **PCs** into the workspace.
- Select the **Network Devices > Routers** category and drag two **Routers** into the workspace.
- Select the **Network Devices > Switches** category and drag two **Switches** into the workspace.
- Connect the devices:
 - Use **Copper straight-through cables** to connect each PC to its respective switch.
 - Use **Copper cross-over cables** to connect each router to its respective switch.
 - **Connect the routers** using a **Serial cable** (for WAN connection between routers).

3. Assign Classful IP Addresses:

- Each router and PC will use a classful IP address.
 - **Router 1:**
 - Click on the router, go to the **Config tab**, and select **FastEthernet0/0**.
 - Assign an IP address such as **192.168.1.1** and the subnet mask **255.255.255.0**.
 - **Router 2:**
 - Click on the router, go to the **Config tab**, and select **FastEthernet0/1**.
 - Assign an IP address such as **192.168.2.1** and the subnet mask **255.255.255.0**.
- **Assign IP Addresses to PCs:**
 - Click on **PC1**, go to the **Desktop tab**, select **IP Configuration**, and set the IP address as **192.168.1.10** with the subnet mask **255.255.255.0**.
 - Do the same for **PC2**, setting the IP address as **192.168.2.10** with the subnet mask **255.255.255.0**.

4. Enable Routing Between Routers:

- To enable communication between the two routers, use **Static Routing**:
 - **Router 1:**
 - Go to **CLI tab**, and enter the following commands:

```
Router> enable
Router# configure terminal
Router(config)# ip route 192.168.2.0 255.255.255.0 [Router2
Interface IP]
```

- **Router 2:**
 - Enter the following commands:

```
Router> enable
Router# configure terminal
Router(config)# ip route 192.168.1.0 255.255.255.0 [Router1
Interface IP]
```

5. Test Connectivity:

- To check if everything is working:
 - Click on **PC1**, open the **Command Prompt** in the **Desktop tab**, and type:


```
ping 192.168.2.10
```
 - If the configuration is correct, you should see replies.

Part 2: Classless Addressing (CIDR)

1. Reuse the Same Topology:

- Use the same topology created in **Part 1**.

2. Assign Classless IP Addresses:

- Classless IP addresses use **Variable Length Subnet Masking (VLSM)**.
- **Router 1:**
 - In **FastEthernet0/0**, set the IP to **192.168.1.1** with a subnet mask **255.255.255.252 (/30)**.
- **Router 2:**
 - In **FastEthernet0/1**, set the IP to **192.168.2.1** with a subnet mask **255.255.255.252 (/30)**.
- **Assign IP Addresses to PCs:**
 - For **PC1**, assign **192.168.1.2** with a subnet mask **255.255.255.252**.
 - For **PC2**, assign **192.168.2.2** with a subnet mask **255.255.255.252**.

3. Enable Routing Between Routers:

- Configure **RIP version 2** for routing between classless networks:
 - **Router 1:**
 - In the **CLI tab**, type:

```
Router> enable
Router# configure terminal
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.1.0
Router(config-router)# exit
```

- **Router 2:**
 - Enter similar commands but for **192.168.2.0** network.

4. Test Connectivity:

- Open the **Command Prompt** on **PC1** and try:

```
ping 192.168.2.2
```

If you see successful replies, the classless addressing configuration works correctly.

Tips:

- **Save your work:** Always save the Packet Tracer file after each major step.
- **Troubleshooting:** If a ping fails, go back to the router or PC configuration and verify IP addresses and routes.